Prompting, Adaptation, and Test-time Repair for English–German Terminology-Aware Translation

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Abstract

Terminology-aware machine translation requires systems to produce fluent translations while also using specified domain terms. This project studies sentence-level English–German terminology-aware translation using the WMT25 terminology setup and Qwen2.5 instruction-tuned models. The experiments compare prompting-based methods, a random terminology control, lightweight LoRA adaptation with cleaner instruction-data formatting, model-size diagnostics, and test-time selection and repair. On 500 development examples with Qwen2.5-1.5B-Instruct, basic terminology prompting improves both BLEU and terminology accuracy over the no-terminology baseline, while stronger prompting increases term accuracy but lowers BLEU. With Qwen2.5-3B-Instruct, this tradeoff becomes less severe, and candidate selection followed by terminology repair gives the strongest automatic result. The random terminology control shows that models may follow supplied terms even when they are not the correct domain terminology. The LoRA results are mixed. The initial setup underperforms and produces chat-style boilerplate, but a cleaner LoRA v2 setup improves over the base model on a held-out split. Overall, simple prompting is a strong baseline, but stronger models and lightweight test-time repair can further improve terminology adherence. The findings also show that dictionary quality and exact-match terminology evaluation require careful handling.

1 Introduction

Machine translation systems are strong on many general translation tasks, but terminology remains difficult in domain-specific settings. In areas such as software, law, medicine, and business, a translation can be fluent but still problematic if it does not use the required technical terms. For example, a system may produce a reasonable German sentence but fail to translate a specific English source term using the target term required by a terminology dictionary.

This project studies sentence-level English–German terminology-aware machine translation using the WMT25 terminology setup (Semenov et al., 2025). The task is not only to translate the source sentence, but also to use the provided German target terms when the corresponding English source terms appear. This makes the task different from ordinary machine translation because the model has to balance general translation quality with explicit lexical constraints.

The report focuses on a small instruction-tuned model setting, using `Qwen/Qwen2.5-1.5B-Instruct` as the main model. This setup is useful because many established terminology-aware MT approaches rely on constrained decoding, dedicated terminology-aware training, or larger systems. Smaller instruction-tuned models are easier to run, but it is less clear how reliable they are when they have to follow terminology constraints. The main question is therefore how far simpler methods can go in this constrained setting: prompting, lightweight adaptation, and test-time refinement.

The experiments are organized around four research questions. RQ1 asks whether inference-time prompting improves terminology use. RQ2 asks whether the provided WMT examples can be reformatted into better terminology-focused instruction-tuning data. RQ3 asks whether lightweight LoRA adaptation improves terminology-aware translation. RQ4 asks whether test-time refinement can improve outputs after generation. To answer these questions, the experiments compare prompting-based methods, a random terminology control, LoRA adaptation, post-processing for chat-style artifacts, model-size diagnostics, and test-time candidate selection and repair.

The results show that terminology prompting is effective, but the best method depends on the model and the metric. For the 1.5B model, basic prompting gives the best balance between BLEU and terminology accuracy among the prompt-only methods, while stronger prompting improves term adherence but lowers BLEU. With the stronger Qwen 3B model, this tradeoff becomes less severe. The strongest automatic result comes from Qwen 3B candidate selection followed by terminology repair, which improves both BLEU and terminology accuracy over the best single prompt.

The adaptation results are more mixed, but still useful. The initial LoRA setup underperforms and produces many chat-style boilerplate artifacts. However, a cleaner LoRA v2 setup improves over the base model on a held-out split, showing that training format matters. Overall, the project makes three main contributions:

it compares several terminology prompting strategies, tests both weak and cleaner LoRA adaptation setups, and shows that simple test-time selection and repair can improve terminology adherence. The results also highlight important risks: noisy dictionaries can mislead the model, stronger terminology control can affect translation quality, and exact-match terminology evaluation can underestimate valid German variants.

2 Related Work

2.1 Terminology-aware Machine Translation

Terminology-aware machine translation studies whether systems can correctly use required domain terms during translation. This is especially important in technical and professional domains, where a fluent translation can still be unsuitable if key terms are translated inconsistently or with the wrong target term. The WMT25 terminology translation task evaluates this problem by giving systems additional terminology information and measuring translation quality, terminology accuracy, and terminology consistency (Semenov et al., 2025). The task includes a sentence-level Track 1 setting for English–German, English–Spanish, and English–Russian, and a document-level Track 2 setting for English–Traditional Chinese and Traditional Chinese–English finance translation (Semenov et al., 2025). This project focuses only on the sentence-level English–German setting.

Earlier work has studied ways to force or encourage specific lexical items in neural generation. Lexically constrained decoding changes the decoding process so that specified terms must appear in the output (Hokamp and Liu, 2017; Post and Vilar, 2018). Another approach is to train NMT systems to apply terminology constraints that are provided at run time (Dinu et al., 2019). These approaches show that explicit terminology information can improve term use, but they also require either modifying the decoding procedure or training the model to handle terminology constraints.

This project does not try to build a new state-of-the-art constrained decoding system. Instead, it studies whether a smaller instruction-tuned model can use terminology through simpler mechanisms that are easier to run in a limited-compute setting. For that reason, prompting is used as the main baseline, while instruction-data formatting, LoRA adaptation, and test-time repair are tested as lightweight extensions.

2.2 Prompting and Instruction-tuned Models

Instruction-tuned language models can often follow natural-language instructions, which makes prompting a simple way to guide translation behavior. In terminology-aware translation, the terminology dictionary can be included directly in the prompt. This avoids changing the model parameters and gives a simple baseline for terminology control. This directly motivates RQ1, which tests whether inference-time prompting strategies can improve term usage.

Prompting also creates a risk. A model may follow supplied terms even when the terminology dictionary is noisy or incorrect. This motivates the random terminology control in this project, where the model receives incorrect term pairs to test whether it follows supplied terminology even when the terminology is wrong. The WMT25 setup also includes no-terminology, proper-terminology, and random-terminology modes, which makes this comparison directly relevant to the task (Semenov et al., 2025).

The experiments use `Qwen2.5-1.5B-Instruct`, a small instruction-tuned model from the Qwen2.5 model family. Qwen2.5 is described as a general-purpose model family with post-training for instruction following, which makes it suitable for testing prompt-based terminology control in a constrained compute setting (Yang et al., 2024). A Qwen2.5-3B diagnostic is also used to check whether the observed prompting tradeoff changes with a stronger model.

2.3 Model Adaptation and Evaluation

Another way to improve terminology behavior is to adapt the model using examples that include term information. Full fine-tuning can be expensive, so parameter-efficient methods such as LoRA are commonly used to adapt large language models with fewer trainable parameters. LoRA freezes the base model and adds trainable low-rank matrices, reducing the cost of adaptation compared with updating all model weights (Hu et al., 2021). This motivates RQ2 and RQ3: the provided WMT examples are reformatted into cleaner terminology-focused instruction-tuning data, and LoRA is used to test whether lightweight adaptation improves terminology-aware translation.

Evaluation is also important in this task. Translation quality is measured with SacreBLEU, which was introduced to make BLEU reporting more standardized and reproducible (Post, 2018). Term use is measured with exact-match terminology accuracy in the main experiments. This metric is direct and easy to interpret, but it can be too strict for morphologically rich languages such as German. Prior work on terminology consistency also argues that terminology evaluation should consider whether terms are translated consistently across contexts, not only whether they appear in one sentence (Semenov and Bojar, 2022). This motivates the relaxed matching and repeated-term consistency diagnostics used later in the report.

3 Method

This project focuses on sentence-level English–German terminology-aware machine translation in Track 1 of the WMT terminology setup (Semenov et al., 2025). The goal is not only to produce a fluent German translation, but also to use the target terms provided in a terminology dictionary. This creates a tradeoff that appears throughout the experiments: a method may improve term usage while still reducing overall translation quality.

The methods are organized around four research questions. RQ1 asks whether inference-time strategies can improve terminology use, so the first set of experiments compares no-term, basic, strong, few-shot, and random terminology prompting. RQ2 asks whether the provided WMT examples can be reformatted to better support terminology-aware instruction tuning, so the examples are converted into cleaner terminology-focused training instances. RQ3 asks whether model adaptation helps, so lightweight LoRA adapters are trained and compared against the base model. RQ4 asks whether an additional advanced strategy can improve the output after generation, so test-time candidate selection and terminology repair are evaluated.

The methods therefore cover four levels of intervention: prompting, random terminology controls, lightweight model adaptation, and test-time refinement. This keeps the comparison focused while still covering inference-time, data-formatting, adaptation, and post-generation strategies.

3.1 Data and Task

The experiments use the English–German development data from the WMT terminology translation task (Semenov et al., 2025). Each example contains an English source sentence, a German reference translation, and term annotations. In the proper terminology setting, the model receives a dictionary of source terms and their required German translations. In the no-term setting, the model only receives the source sentence.

A random terminology setting is also used as a control condition. Here, the model receives term pairs that are not the correct domain-specific terms for the sentence. This is not meant to be a useful translation strategy. Instead, it tests whether the model follows supplied terms even when those terms are wrong.

No external parallel or monolingual data is added. For the training-data modification part of the project, the provided WMT examples are reformatted into instruction-tuning examples. Each example contains the source sentence and the term dictionary as input, while the German reference translation is used as the target output. In the cleaner LoRA v2 setup, the target completion is always only the German translation, without chat-style prefixes or explanations.

3.2 Models

The main experiments use `Qwen/Qwen2.5-1.5B-Instruct`, an instruction-tuned model from the Qwen2.5 family (Yang et al., 2024). This model was chosen because it is small enough to run with the available GPU resources, while still being instruction-tuned and suitable for prompt-following experiments. Inputs are formatted using the model's chat template to match the instruction-tuned setup. During evaluation, decoding is deterministic so that differences between methods are easier to compare.

After the main experiments, a model-size diagnostic is also run with `Qwen/Qwen2.5-3B-Instruct` (Yang et al., 2024). This diagnostic tests whether the quality–adherence tradeoff observed with the 1.5B model changes with a stronger instruction-tuned model. The 3B model is not treated as a replacement for the main setup, but as an additional comparison.

3.3 Prompting Strategies

Several prompting strategies are evaluated. The no-term baseline asks the model to translate the source sentence without providing any dictionary. The basic prompt gives the source sentence and the term list, with an instruction to use the provided terms when relevant. The strong prompt uses a stricter instruction and tells the model to use the exact target terms whenever the corresponding source terms appear. A few-shot prompt is also tested, where example translations are included before the test sentence. Finally, the random condition uses the same basic prompt format, but with random rather than proper terms.

Condition	Description
No-term	Source only.
Basic	Source + term dictionary.
Strong	Source + stricter exact-term instruction.
Few-shot	Adds example translations.
Random	Uses incorrect terms as a control.

Table 1: Prompting conditions used in the experiments.

3.4 LoRA Adaptation

To test whether model adaptation improves terminology-aware translation, lightweight LoRA adapters are trained on reformatted instruction-tuning examples. LoRA is used as a parameter-efficient fine-tuning method, so only a small set of adapter parameters is trained while the base model remains frozen (Hu et al., 2021).

An initial LoRA adapter is first trained on terminology-aware instruction examples. The adapted model is evaluated in two ways. First, it is tested without a terminology prompt to check whether adaptation alone teaches the model to use the required terms. Second, it is tested with the same basic prompt used for the base model, to check whether LoRA and inference-time prompting are complementary.

After weak initial LoRA results, a cleaner LoRA v2 adapter is trained. This setup connects RQ2 and RQ3: the adaptation method is still LoRA, but the training data is cleaned and formatted more carefully. Each training example is converted into two prompt variants, one with a basic terminology instruction and one with a stronger exact-term instruction. The target completion is always the German reference translation only. This version tests whether the initial LoRA failure was partly caused by the adaptation format rather than by LoRA itself.

A small LoRA training-size ablation is also run using

50, 100, 200, and 400 source examples. All adapters are evaluated on the same held-out 100-example split. This tests whether increasing the amount of clean terminology-focused adaptation data improves LoRA performance.

3.5 Post-processing and Test-time Refinement

During qualitative inspection, the initial LoRA-adapted model often produced chat-style boilerplate, such as *Gerne! Hier ist die deutsche Übersetzung:*. Because this type of prefix can hurt automatic translation metrics, a simple cleanup step is added to remove common translation-introduction phrases. This cleanup is not a new translation system, but a diagnostic post-processing step.

Terminology repair is also tested as a lightweight test-time refinement method. For outputs that are missing required target terms, the model is prompted to revise the translation while keeping the original source meaning and inserting the missing terminology. This repair step is applied to selected Qwen 3B outputs and to LoRA outputs as a way to test whether missing terms can be fixed after generation.

Finally, for the Qwen 3B diagnostic, a simple candidate selection strategy is tested. For each source sentence, the selector chooses among the no-term, basic, and strong prompt outputs based on terminology coverage, using prompt strength as a tie-breaker. The selector is a lightweight rule-based method rather than a learned reranker. It is included as the RQ4 method because it tests whether terminology-aware MT can be improved after generation, especially when different prompts succeed on different examples.

4 Experimental Setup

The main evaluation uses 500 examples from the English–German development data. This subset is used for the main prompting, random terminology, initial LoRA, cleanup, Qwen 3B, candidate selection, terminology repair, relaxed matching, and repeated-term consistency experiments. Smaller runs were used during development to debug the pipeline and test early prompt variants. A 100-example subset is kept for additional diagnostics, including few-shot prompting, prompt-refinement tests, the clean LoRA v2 experiment, the LoRA training-size ablation, and the model-family comparison.

The experimental conditions are set up as controlled comparisons. No-term versus basic prompting tests the effect of adding terminology. Basic versus strong prompting tests whether stricter instructions improve term use. The random terminology condition tests whether the model follows supplied terminology even when the dictionary is wrong. The LoRA conditions test lightweight adaptation, while cleanup and repair test whether artifacts or missing terms can be fixed after generation. The Qwen 3B diagnostic tests whether the quality–adherence tradeoff changes with a stronger base model.

Translation quality is evaluated with BLEU using SacreBLEU (Post, 2018). Terminology use is evaluated with exact-match terminology accuracy. For each required target term, the evaluation checks whether that target term appears in the generated German output. The final score is the proportion of required target terms that appear across all evaluated examples.

For the random terminology condition, two terminology scores are reported. The first measures adherence to the supplied random terms. The second measures adherence to the proper terms for the same examples. This distinction matters because a model can follow the supplied random dictionary while failing to use the correct domain terminology.

All generations use deterministic decoding. The maximum number of new tokens is set to 160. The main Qwen 1.5B experiments use a batch size of 2. Later Qwen 3B and diagnostic runs use small batch sizes that fit the available GPU memory. To avoid mixing base and adapted systems, the base model and LoRA-adapted model are stored separately and passed explicitly to the generation function. An earlier unverified LoRA run was discarded after identical base and LoRA outputs showed that the global model variable had been overwritten.

The main 500-example systems are no terminology, basic prompting, strong prompting, random terminology, initial LoRA without terminology prompting, initial LoRA with basic prompting, and initial LoRA with basic prompting followed by cleanup. The Qwen 3B diagnostic is evaluated on the same 500-example subset in the no-term, basic, and strong prompting settings. For Qwen 3B, candidate selection chooses among the no-term, basic, and strong outputs for each example, and terminology repair is applied to outputs with missing terms.

The clean LoRA v2 diagnostic uses a separate 400/100 split of the 500-example subset. The first 400 examples are used for adaptation, and the remaining 100 examples are held out for evaluation. This setup tests whether reformatting the provided WMT examples into cleaner translation-only instruction data improves LoRA adaptation. A small LoRA training-size ablation is also run on the same held-out 100 examples, using 50, 100, 200, and 400 training examples.

Few-shot prompting is evaluated only on a 100-example subset, since early results showed weaker performance than simpler prompting strategies. Prompt-refinement diagnostics and the model-family comparison are also kept as 100-example diagnostics rather than main results. These experiments are used to understand specific failure modes, not to replace the main 500-example evaluation.

To complement sentence-level terminology accuracy, a lightweight repeated-term consistency diagnostic is also reported. For each source–target term pair that appears more than once in the 500-example subset, the diagnostic measures how often each method produces the required target term. This is not the official WMT terminology consistency metric. An attempt was made to run the official WMT term-consistency script, but the available implementation depends on an LLM-based term-alignment step requiring a valid OpenAI API key. The repeated-term diagnostic is therefore used as a simpler consistency check.

5 Results

The main results are reported on 500 English–German development examples. These results compare the main prompting conditions, the random terminology control, the initial LoRA setup, and boilerplate cleanup.

Method	BLEU	Term Accuracy
No terminology	19.83	0.262
Basic prompting	21.63	0.700
Strong prompting	17.91	0.741
Random terminology	21.28	0.746 / 0.296
LoRA no-term	11.47	0.189
LoRA + basic prompt	12.46	0.628
LoRA + basic + cleanup	14.17	0.627

Table 2: Main EN–DE terminology translation results on 500 development examples. For the random terminology condition, terminology accuracy is reported as random-term adherence / proper-term adherence.

Table 2 gives the first answer to RQ1: terminology prompting clearly improves term use compared with the no-terminology baseline. The no-terminology baseline reaches 19.83 BLEU and 0.262 terminology accuracy, so the model sometimes produces the required terms without guidance, but not reliably.

Basic prompting improves both metrics. BLEU increases from 19.83 to 21.63, and terminology accuracy increases from 0.262 to 0.700. Among the prompt-only methods for the 1.5B model, this gives the best balance between translation quality and terminology adherence. Strong prompting reaches higher terminology accuracy at 0.741, but BLEU drops to 17.91. This supports the quality–adherence tradeoff tested in RQ1: stricter term instructions can improve lexical matching, but they can also reduce translation quality.

The random terminology control behaves differently from the proper terminology settings. It reaches 21.28 BLEU and 0.746 adherence to the supplied random terms. However, when the same outputs are evaluated against the proper terms, terminology accuracy is only 0.296. The model is therefore sensitive to terms supplied in the prompt, but it does not reliably distinguish correct domain terminology from incorrect terminology. For that reason, random terminology is treated as a control condition rather than a useful translation method.

The initial LoRA results answer the first part of RQ3. Without a terminology prompt, the LoRA-adapted model performs worse than the no-terminology base model, reaching 11.47 BLEU and 0.189 terminology accuracy. With the basic prompt, LoRA recovers some terminology adherence, reaching 0.628, but BLEU remains low at 12.46. This initial LoRA setup therefore does not improve terminology-aware translation.

Boilerplate cleanup partly improves the initial LoRA output. Removing common chat-style prefixes increases BLEU from 12.46 to 14.17, while terminology accuracy remains almost unchanged. Boilerplate artifacts contributed to the low BLEU score, but they do not fully explain the gap between the initial LoRA setup and the base prompting methods.

5.1 Clean LoRA v2 Diagnostic

After the initial LoRA setup underperformed, a cleaner LoRA v2 diagnostic was run to test whether the adaptation format mattered. The 500-example subset was split into 400 training examples and 100 held-out evaluation examples. Each training example was converted into two instruction-tuning variants: one with a basic terminology prompt and one with a stronger exact-term prompt. The target completion was always only the German reference translation, without chat-style prefixes.

The full LoRA v2 diagnostic table is reported in Appendix A. Unlike the initial LoRA setup, the cleaner LoRA v2 adapter improves over the base model on the same held-out split. LoRA v2 with the basic prompt reaches 23.24 BLEU and 0.773 terminology accuracy, compared with 17.56 BLEU and 0.709 terminology accuracy for the base model with the same prompt. Applying terminology repair to the LoRA v2 basic outputs improves the result further to 24.14 BLEU and 0.882 terminology accuracy. This indicates that the adaptation format matters: cleaner translation-only targets and terminology-focused prompt variants make LoRA more useful in this setup.

A small training-size ablation is reported in Appendix D. It shows that BLEU improves with more clean terminology-focused adaptation examples, while terminology accuracy is less monotonic in the smaller settings.

5.2 Few-shot Prompting

Few-shot prompting was evaluated on the 100-example subset as an additional inference-time comparison. It achieved 14.15 BLEU and 0.560 terminology accuracy, which is lower than both basic and strong prompting on the same subset. In this setup, adding in-context examples did not help the smaller instruction-tuned model. A likely reason is that the longer prompt made the task more complex instead of making the terminology constraint clearer.

5.3 Model-size Diagnostic

After the main Qwen 1.5B experiments, an additional 500-example diagnostic was run with `Qwen/Qwen2.5-3B-Instruct`. This tests whether the quality–adherence tradeoff changes with a stronger instruction-tuned model.

The 3B no-terminology setting reaches higher BLEU than the 1.5B no-terminology baseline, but terminology accuracy remains low at 0.287. The larger model

Method	BLEU	Term Acc.	Boilerplate
1.5B no-term	19.83	0.262	6
1.5B basic	21.63	0.700	3
1.5B strong	17.91	0.741	4
3B no-term	22.93	0.287	1
3B basic	20.37	0.898	0
3B strong	24.58	0.876	0

Table 3: Additional 500-example Qwen model-size diagnostic.

improves general translation quality, but still does not reliably produce the required terms without terminology guidance. With terminology prompting, the 3B model improves term accuracy strongly. The 3B basic prompt reaches 0.898 terminology accuracy, while the 3B strong prompt gives the best single-prompt result, with 24.58 BLEU and 0.876 terminology accuracy. This result qualifies the RQ1 tradeoff: the prompt matters, but the capability of the base model also matters.

5.4 Candidate Selection and Terminology Repair

As the main RQ4 experiment, Qwen 3B outputs were used for candidate selection and terminology repair. First, the system selected among the no-term, basic, and strong prompting outputs for each sentence based on exact terminology coverage, using prompt strength as a tie-breaker. Second, outputs that were still missing required terms were passed through a repair prompt that asked the model to revise the translation while preserving meaning and adding the missing terms.

Method	BLEU	Term Acc.	Boilerplate
3B strong	24.58	0.876	0
3B selected	24.89	0.928	0
3B selected + repair	25.05	0.968	0

Table 4: Qwen 3B candidate selection and terminology repair results on 500 examples.

The selector chose the strong-prompt output for 473 examples, the basic-prompt output for 24 examples, and the no-term output for 3 examples. Candidate selection improves over the best single prompt, increasing BLEU from 24.58 to 24.89 and terminology accuracy from 0.876 to 0.928. The repair step improves terminology accuracy further to 0.968 and slightly increases BLEU to 25.05. Before repair, 33 outputs were still missing at least one required term. After repair, this decreased to 15 outputs. This is the strongest automatic result in the project and shows that a simple test-time selection and repair pipeline can improve both translation quality and terminology adherence without extra training.

5.5 Repeated-term Consistency Diagnostic

To complement sentence-level terminology accuracy, a lightweight repeated-term consistency diagnostic was also run. For each source–target term pair that appeared more than once in the 500-example subset, the diagnostic measures how often each method produced the required target term. This is not the official WMT terminology consistency metric, but it gives a simple view of whether repeated terms are handled reliably across examples.

The full consistency table is reported in Appendix G. The main trend matches the sentence-level results: Qwen 3B methods are more consistent than the 1.5B prompting baselines, candidate selection improves consistency further, and the selected + repair system gives the highest scores. It reaches 0.957 average repeated-term success and 0.909 perfect consistency. This suggests that targeted terminology repair improves not only overall terminology accuracy, but also reliability for terms that appear repeatedly.

6 Discussion

The results show that terminology-aware translation is not a case where one method wins on every metric. The main pattern is a tradeoff between general translation quality and explicit term use. For the 1.5B model, basic prompting improves both BLEU and terminology accuracy over the no-terminology baseline, while strong prompting gives higher term accuracy but lower BLEU. Stricter terminology instructions can therefore push the model toward lexical matching, but this does not automatically produce better translations.

6.1 Prompting and the Quality–Adherence Tradeoff

For the 1.5B model, the strongest prompt-only balance comes from basic prompting. It improves BLEU from 19.83 to 21.63 and terminology accuracy from 0.262 to 0.700. Strong prompting increases terminology accuracy further to 0.741, but BLEU drops to 17.91. This supports the main tradeoff in the project: forcing exact terms can improve lexical adherence, but it can also make the output less natural or less close to the reference.

This result is consistent with earlier terminology-aware MT work in the sense that terminology constraints can improve required term use, but they still need to be integrated carefully into the translation process (Dinu et al., 2019). The difference in this project is that the constraint is mainly given through prompting, rather than through a dedicated constrained decoding algorithm or a fully trained terminology-aware NMT system (Hokamp and Liu, 2017; Post and Vilar, 2018). This makes prompting easy to apply, but also less controlled.

The Qwen 3B diagnostic changes the picture. With the 1.5B model, stronger prompting improves term accuracy but hurts BLEU. With the 3B model, strong prompting achieves both high BLEU and high terminology accuracy. At the same time, the 3B no-terminology setting still has low terminology accuracy despite strong BLEU. The tradeoff is therefore not only caused by the prompt. The capability of the base model also matters, but terminology guidance is still needed. Additional 100-example model-family comparisons are reported in

Appendix B.

The candidate selection and repair experiment gives the strongest test-time result. Instead of relying on one fixed prompt, the system selects among multiple prompt outputs for each sentence. This improves both BLEU and terminology accuracy for Qwen 3B. The repair step then reduces the number of outputs with missing terms. The repeated-term diagnostic supports the same conclusion: selection and repair improve not only sentence-level terminology accuracy, but also reliability for terms that appear multiple times. This suggests that lightweight selection and revision are useful directions for terminology-aware MT. At the same time, repair explicitly optimizes for missing terms, so human evaluation would still be needed to check whether revised outputs remain fluent and adequate.

A more verbose definition-style terminology prompt, reported in Appendix C, did not improve over the simpler strong prompt or the selection-and-repair system. This suggests that more detailed terminology formatting is not automatically better.

6.2 Random Terminology as a Control

The random terminology condition shows an important risk of prompting. The 1.5B model follows the supplied random terms with high accuracy, reaching 0.746 random-term adherence. However, when the same outputs are evaluated against the proper terms, terminology accuracy drops to 0.296, which is close to the no-terminology baseline.

The additional Qwen 3B random-terminology experiment strengthens this point. Even with the stronger 3B model, random terminology prompting leads to high adherence to the supplied random terms at 0.752, while proper-term accuracy drops to 0.167 and BLEU falls to 10.84. Stronger model capacity therefore does not automatically protect the system from an incorrect dictionary. If the supplied terminology is wrong, the model may still follow it and produce worse translations.

This shows both the usefulness and the risk of terminology prompting. A correct dictionary can guide the model toward better term use. A noisy or incorrect dictionary can mislead it. Dictionary quality is therefore part of the translation system, not just an external input. This is especially important for real terminology-aware MT settings, where term lists may come from user-provided dictionaries or domain resources and may not always be clean.

6.3 LoRA Adaptation and Chat-style Artifacts

The initial LoRA results were weaker than expected. LoRA without a terminology prompt performs worse than the base no-terminology model, and LoRA with the basic prompt does not match the base prompting methods. This first LoRA setup was not enough to teach the model better terminology-aware translation behavior.

The cleaner LoRA v2 diagnostic makes the LoRA result less straightforward. This also addresses RQ2, since the improvement comes from changing how the provided WMT examples are formatted for instruction tuning rather than from adding external training data. After the training data was reformatted into cleaner translation-only instruction examples, LoRA improved over the base model on a held-out 100-example split. LoRA v2 with the basic prompt reaches 23.24 BLEU and 0.773 terminology accuracy, compared with 17.56 BLEU and 0.709 terminology accuracy for the base model with the same prompt. Applying terminology repair to the LoRA v2 basic outputs improves the result further to 24.14 BLEU and 0.882 terminology accuracy. The poor initial LoRA result was therefore partly related to the adaptation setup, not necessarily to LoRA itself. Still, because LoRA v2 is evaluated mainly as a smaller held-out diagnostic, the main conclusions rely on the 500-example prompting and Qwen 3B experiments.

This fits the motivation for parameter-efficient adaptation: LoRA can change model behavior without full fine-tuning, but the training format still matters (Hu et al., 2021). In this project, cleaner target completions reduced the chat-style artifact problem and made adaptation more useful. The training-size ablation in Appendix D supports the same point: more clean adaptation data improves BLEU, although terminology accuracy is less monotonic in the smaller settings.

A qualitative inspection helped explain part of the initial LoRA failure. The LoRA-adapted model often produced chat-style boilerplate, such as *Gerne! Hier ist die deutsche Übersetzung:*. This kind of output is understandable for an instruction-tuned chat model, but it is undesirable in machine translation evaluation because the expected output is only the translation. Boilerplate patterns are rare in the base prompting outputs but common in the initial LoRA outputs, especially in the LoRA + basic prompt condition. The full boilerplate counts are reported in Appendix E.

These counts are based on simple substring matching, so they should be read as diagnostic evidence rather than a full manual error annotation. Still, the difference between base prompting and initial LoRA is large enough to show a clear pattern.

The cleanup experiment confirms that these artifacts affected the automatic scores. Removing common boilerplate prefixes changes 95 LoRA + basic outputs and improves BLEU from 12.46 to 14.17, while terminology accuracy remains almost unchanged. However, cleaned initial LoRA still remains far below base basic prompting. Boilerplate partly explains the poor initial LoRA BLEU score, but it does not fully explain the underperformance. The adapted model also appears to produce lower-quality translations more generally.

Terminology repair can partially rescue weak adapted outputs. Applied to the cleaned LoRA outputs, repair improves BLEU from 14.17 to 19.27 and terminology accuracy from 0.627 to 0.928. The number of outputs with missing terms decreases from 193 to 38. However, repaired LoRA still remains below the Qwen 3B selected + repair system, which reaches 25.05 BLEU and 0.968 terminology accuracy with no boilerplate. Repair helps, but it does not fully compensate for a weak adapted model.

Two smaller prompt-refinement diagnostics were also tested to reduce chat-style artifacts. A stricter translation-only prompt reduced boilerplate to 2 outputs, but did not improve over basic prompting: BLEU was 18.40 and terminology accuracy was 0.690, compared with 20.86 BLEU and 0.698 terminology accuracy for basic prompting. A lighter output-only instruction also kept boilerplate low, but still underperformed the original basic prompt. Output-format instructions can reduce chat-style artifacts, but they do not necessarily improve translation quality or terminology adherence.

6.4 Qualitative Error Patterns

The qualitative examples show why the automatic scores should be interpreted carefully. In one example, the required term was *check* → *Kontrolle*. Some outputs used related forms such as *Kontrollieren* or *Kontrolliere*. These forms are connected to the required term, but they are not counted by exact matching. This is one reason why example inspection is still useful.

A second pattern appears with terms such as *provider* → *Provider*. Prompting often encouraged the model to include the required term, but the surrounding German could still be awkward. This matches the quantitative tradeoff: inserting the correct lexical item does not guarantee a fluent or natural translation.

The clearest example is *consumption model* → *Verbrauchsmodell*. Basic prompting successfully produced the required term, while the initial LoRA + basic output also used the term but added a chat-style prefix such as *Gerne! Hier ist die deutsche Übersetzung:*. This example links three recurring issues in the project: terminology success, instruction-following behavior, and the boilerplate problem.

6.5 Limitations of Exact-match Term Accuracy

Exact-match terminology accuracy is useful because it directly checks whether the required target terms appear in the output. However, terminology evaluation is not only about sentence-level matching, since consistency across repeated terms also matters (Semenov and Bojar, 2022). Exact matching can also be too strict for German. Some outputs use morphologically related or inflected forms that may be acceptable in context but are not counted as exact matches. For example, *Kontrolle*, *Kontrollieren*, and *Kontrolliere* are related, but the metric treats them differently.

A relaxed matching diagnostic was used to check this issue. The relaxed score increases terminology accuracy for all methods, especially strong prompting, which rises from 0.741 to 0.810. This relaxed metric is not used as the main result because it is heuristic and may overcount some incorrect matches. Still, it supports the observation that exact matching may underestimate valid term use in German. The full exact-versus-relaxed matching table is reported in Appendix F.

Overall, the results show that terminology control improves lexical adherence, but reliable terminology-aware MT also depends on prompt design, dictionary quality, output formatting, model capability, and evaluation choices.

7 Conclusion

This project evaluated prompting, lightweight LoRA adaptation, and test-time refinement for English–German terminology-aware machine translation. The main finding is that terminology prompting is effective, but the best method depends on the metric and the model. For the 1.5B model, basic prompting gave the best balance among the prompt-only methods, while strong prompting improved terminology adherence at the cost of lower BLEU. With the stronger Qwen 3B model, this tradeoff became less severe, and strong prompting achieved both high BLEU and high terminology accuracy.

The random terminology control showed that the model is sensitive to terms supplied in the prompt, even when those terms are not the correct domain terminology. This makes dictionary quality an important part of the translation pipeline. If the supplied terminology is wrong, the model may still follow it and produce worse translations.

The LoRA results were mixed. The initial LoRA setup underperformed and produced many chat-style boilerplate artifacts. However, the cleaner LoRA v2 setup improved over the base model on a held-out split. This shows that LoRA can help in this task, but only when the provided WMT examples are formatted carefully as instruction-tuning data. It also suggests that the first LoRA failure was not only an adaptation problem, but also a data-formatting problem.

The strongest automatic results came from test-time refinement with Qwen 3B. Candidate selection improved over the best single prompt, and terminology repair further increased terminology accuracy while also slightly improving BLEU. The repeated-term diagnostic showed the same pattern, with the selected + repair system giving the most reliable repeated-term behavior.

These findings are consistent with earlier terminology-aware MT work showing that explicit terminology constraints can improve required term use, but they also show that prompting-based terminology control has its own risks. Unlike constrained decoding or fully trained terminology-aware MT systems, prompting is easy to apply but depends strongly on dictionary quality, model capability, and output formatting.

Overall, simple prompting is a strong baseline, but the best performance comes from combining a stronger model with terminology-aware selection and repair. Future work should test official submissions, more language pairs, larger adapted models, and human evaluation, especially because exact matching can miss valid German inflections or related forms.

Limitations

This project has several limitations. First, the experiments focus only on sentence-level English–German translation. Other Track 1 language pairs, document-level Track 2, and general MT test sets such as FLORES or WMT24++ are not evaluated. This keeps the setup controlled, but it also limits how far the findings can be generalized.

Second, the experiments use the provided WMT terminology development data rather than an official test submission. This is useful for comparing methods under the same conditions, but the results should be interpreted as development-set findings rather than shared-task leaderboard results. Because the experiments were developed iteratively on this subset, some choices may also be tuned to this development setting.

Third, the main automatic metrics are BLEU and exact-match terminology accuracy. BLEU does not fully capture translation adequacy or fluency, and exact-match terminology accuracy can miss valid German variants. This issue is partly addressed with a relaxed matching diagnostic and qualitative analysis, but a more complete evaluation would include human judgments or neural metrics such as COMET or MetricX.

Fourth, the LoRA and training-data modification experiments are still small-scale. The initial LoRA setup underperformed, while the cleaner LoRA v2 setup improved over the base model on a held-out split. However, the training-data modification is limited to reformatting the provided WMT examples into cleaner instruction-tuning data. The adaptation experiments also do not include extensive hyperparameter tuning, external training data, or larger adapted models. The results therefore show that LoRA is sensitive to training format and data setup, not that this is the strongest possible LoRA configuration.

Fifth, the test-time repair step is evaluated automatically. Repair improves terminology accuracy and repeated-term consistency, but it explicitly optimizes for inserting missing terms. A human evaluation would be needed to check whether repaired translations always preserve meaning and remain natural German. This is especially important because a repair step can improve lexical matching while still making the sentence awkward or changing the meaning.

Finally, the official WMT terminology consistency metric is not reported as a main result. An attempt was made to run the official term-consistency script, but the available implementation depends on an LLM-based term-alignment step requiring a valid OpenAI API key. For this reason, the report uses a lightweight repeated-term consistency diagnostic instead. This diagnostic is less comprehensive than the official metric, but it still gives useful evidence about repeated terminology reliability.

Future work could evaluate additional language pairs, run an official shared-task submission, test stronger adaptation setups, use human evaluation, and measure terminology consistency across larger datasets or document-level settings.

Contributions

Earlier project planning and the midterm project stage, including initial discussion of research directions, were discussed jointly by Annabelle Donato and Li Tian. For the final project submission, Annabelle Donato was responsible for the experimental design, implementation of the prompting experiments, LoRA adaptation experiments, model-size diagnostics, test-time selection and repair experiments, evaluation scripts, result analysis, report writing, and repository organization.

Code Availability

The project repository contains the notebooks, result files, final report PDF, and reproducibility materials: `https://github.com/bLacKr0sE17/nlp2-terminology-translation-final`.

References

Loubna Ben Allal, Anton Lozhkov, Elie Bakouch, Gabriel Martín Blázquez, Guilherme Penedo, Lewis Tunstall, Andrés Marafioti, Hynek Kydlíček, Agustín Piqueres Lajarín, Vaibhav Srivastav, Joshua Lochner, Caleb Fahlgren, Xuan-Son Nguyen, Clémentine Fourrier, Ben Burtenshaw, Hugo Larcher, Haojun Zhao, Cyril Zakka, Mathieu Morlon, and 3 others. 2025. SmolLM2: When Smol Goes Big – Data-Centric Training of a Small Language Model. *arXiv preprint arXiv:2502.02737*.

Georgiana Dinu, Prashant Mathur, Marcello Federico, and Yaser Al-Onaizan. 2019. Training Neural Machine Translation to Apply Terminology Constraints. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3063–3068, Florence, Italy. Association for Computational Linguistics.

Chris Hokamp and Qun Liu. 2017. Lexically Constrained Decoding for Sequence Generation Using Grid Beam Search. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1535–1546, Vancouver, Canada. Association for Computational Linguistics.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. LoRA: Low-Rank Adaptation of Large Language Models. *arXiv preprint arXiv:2106.09685*.

Matt Post. 2018. A Call for Clarity in Reporting BLEU Scores. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pages 186–191, Brussels, Belgium. Association for Computational Linguistics.

Matt Post and David Vilar. 2018. Fast Lexically Constrained Decoding with Dynamic Beam Allocation for Neural Machine Translation. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 1314–1324, New Orleans, Louisiana. Association for Computational Linguistics.

Kirill Semenov and Ondřej Bojar. 2022. Automated Evaluation Metric for Terminology Consistency in MT. In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 450–457, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics.

Kirill Semenov, Xu Huang, Vilém Zouhar, Nathaniel Berger, Dawei Zhu, Arturo Oncevay, and Pinzhen Chen. 2025. Findings of the WMT25 Terminology Translation Task: Terminology is Useful Especially for Good MTs. In *Proceedings of the Tenth Conference on Machine Translation*, pages 554–576, Suzhou, China. Association for Computational Linguistics.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, and 9 others. 2025. Qwen3 Technical Report. *arXiv preprint arXiv:2505.09388*.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, and 23 others. 2024. Qwen2.5 Technical Report. *arXiv preprint arXiv:2412.15115*.

Appendix

A Clean LoRA v2 Diagnostic

A cleaner LoRA v2 diagnostic was run after the initial LoRA setup underperformed. The adapter was trained on 400 examples and evaluated on a held-out 100-example split.

Method	BLEU	Term Acc.	BP
Base basic	17.56	0.709	0
LoRA v2 basic	23.24	0.773	0
LoRA v2 strong	21.38	0.773	0
LoRA v2 basic + repair	24.14	0.882	0

Table 5: Clean LoRA v2 diagnostic on a held-out 100-example split. BP denotes boilerplate count.

B Additional Model-family Comparison

An additional 100-example comparison was run using basic terminology prompting across several small instruction-tuned models, including Qwen3 and SmolLM2 variants (Yang et al., 2025; Allal et al., 2025). This experiment was used as a diagnostic analysis rather than as a main result.

Model / Method	BLEU	Term Acc.	BP
Qwen2.5-0.5B basic	6.28	0.638	23
Qwen2.5-1.5B basic	20.86	0.698	1
Qwen3-1.7B clean	24.51	0.767	0
Qwen2.5-3B basic	20.62	0.888	0
SmolLM2-1.7B basic	6.73	0.741	0

Table 6: Additional 100-example model-family comparison using basic terminology prompting. BP denotes boilerplate count.

This comparison suggests that terminology prompting is highly model-dependent. Smaller or weaker models sometimes followed terminology but produced low-quality translations, while stronger Qwen models gave better quality–adherence tradeoffs. The Qwen3 result also shows that prompt formatting matters: a clean output-only prompt was needed to obtain usable translations.

C Additional Prompt-format Diagnostic

A more verbose definition-style terminology prompt was tested on the 500-example subset to check whether providing more detailed term information improves terminology-aware translation.

Method	BLEU	Term Acc.	Boilerplate
3B basic	20.37	0.898	0
3B strong	24.58	0.876	0
3B selected + repair	25.05	0.968	0
3B definition prompt	22.58	0.880	2

Table 7: Additional 500-example prompt-format diagnostic using a definition-style terminology prompt.

The definition-style prompt improved BLEU over the 3B basic prompt, but it did not outperform the simpler strong prompt or the selection-and-repair system. This suggests that more verbose terminology formatting is not automatically better. In this setup, the strongest gains came from test-time selection and targeted repair rather than from adding more detailed terminology descriptions to the prompt.

D LoRA Training-size Ablation

A small training-size ablation was run for the cleaner LoRA v2 setup. All adapters were evaluated on the same held-out 100-example split using the basic terminology prompt.

Training examples	BLEU	Term Acc.	Boilerplate
50	19.35	0.727	0
100	19.85	0.700	0
200	20.73	0.673	0
400	23.24	0.782	0
400 + repair	23.86	0.864	0

Table 8: LoRA v2 training-size ablation on the held-out 100-example split.

The results show that BLEU improves as the amount of clean terminology-focused adaptation data increases. Terminology accuracy is less monotonic for the smaller adapters, but the 400-example adapter gives the strongest unrepaired result. Applying terminology repair to the 400-example adapter further improves terminology accuracy from 0.782 to 0.864.

E Boilerplate Diagnostic

A simple substring-based diagnostic was used to count chat-style boilerplate patterns in the 500-example outputs.

Method	Boilerplate
No terminology	6
Basic prompting	3
Strong prompting	4
Random terminology	6
LoRA no-term	127
LoRA + basic prompt	157

Table 9: Boilerplate pattern counts in 500 EN–DE generations. Counts are based on substring matching.

F Exact-match and Relaxed Terminology Matching

A relaxed matching diagnostic was used to check whether exact-match terminology accuracy undercounts related German forms.

G Repeated-term Consistency Diagnostic

A lightweight repeated-term consistency diagnostic was used to check whether methods handle recurring source–target term pairs reliably across the 500-example subset.

Method	Exact	Relaxed
No terminology	0.262	0.273
Basic prompting	0.700	0.741
Strong prompting	0.741	0.810
LoRA + basic prompt	0.628	0.636

Table 10: Diagnostic comparison of exact-match and relaxed terminology accuracy on 500 EN–DE examples. Relaxed matching is used only as an analysis tool.

The diagnostic includes 88 repeated source–target term pairs.

Method	Avg. success	Perfect cons.
1.5B basic	0.726	0.511
1.5B strong	0.785	0.591
3B basic	0.889	0.716
3B strong	0.891	0.761
3B selected	0.919	0.807
3B selected + repair	0.957	0.909

Table 11: Repeated-term consistency diagnostic on 500 examples. Avg. success is the average repeated-term success rate, and perfect cons. is the share of repeated term pairs translated consistently in all occurrences.